0.0.1 Annulus and spherical-shell Stokes benchmarks

The Cartesian benchmarks provide a useful verification framework for Stokes flow on simple planar domains. To further assess the robustness of the solver in geometries relevant to geodynamics, we also consider annulus and spherical-shell benchmark problems. These benchmarks introduce additional complexities associated with curved boundaries, radial gravity, non-trivial pressure distributions, and velocity boundary conditions imposed on non-planar surfaces. Such configurations are representative of mantle convection and lithospheric deformation models, and therefore provide a more stringent test of the finite-element implementation.

Model accuracy is quantified using relative volume L_2-norm errors for the velocity and pressure fields. For a numerical solution q_h and analytical reference solution q^*, the relative error is defined as

$$E_{L_2}^{\mathrm{rel}}(q) = \left(\frac{\int_\Omega |q_h - q^*|^2 \,\mathrm{d}\Omega}{\int_\Omega |q^*|^2 \,\mathrm{d}\Omega} \right)^{1/2}. \tag{1}$$

This definition is applied to both velocity and pressure. Since the incompressible Stokes equations determine pressure only up to an additive constant, the numerical and analytical pressures are compared after removing their mean values to enforce a common pressure gauge.

Benchmark setup The annulus benchmarks use analytical Stokes solutions defined in cylindrical shell geometry. The Thieulot–Puckett annulus benchmark provides a smooth manufactured solution for incompressible Stokes flow driven by radial gravity, with tangential velocity conditions imposed on the inner and outer boundaries. This benchmark is well suited for assessing the optimal convergence behaviour of mixed finite-element discretisations [Thieulot and Puckett, 2018, Girault and Raviart, 1986, Boffi et al., 2013]. The Kramer annulus benchmark extends this framework by distinguishing between smooth volumetric forcing and delta-function forcing applied on an internal circular interface, and by considering both free-slip and no-slip boundary conditions [Kramer et al., 2021].

The spherical-shell benchmarks provide analogous tests in three-dimensional spherical geometry. The Thieulot spherical benchmark examines smooth Stokes flow in a spherical shell for both constant-viscosity and radially varying viscosity configurations [Thieulot, 2017]. The Kramer spherical benchmark again separates smooth volumetric forcing from delta-function forcing on an internal spherical interface [Kramer et al., 2021]. This distinction is important because the smooth cases primarily test the approximation properties and convergence rates of the finite-element spaces, whereas the delta-function cases introduce reduced solution regularity and therefore provide a more challenging assessment of the numerical method.

Results The smooth Thieulot–Puckett annulus benchmark reproduces the expected convergence hierarchy for mixed finite-element discretisations. The Taylor–Hood $P_2 \times P_1$ discretisation achieves approximately third-order convergence in velocity and second-order convergence in pressure in the volume L_2 norm. Increasing the polynomial order to the $P_3 \times P_2$ pair further improves the convergence rates to approximately fourth order in velocity and third order in pressure before the finest meshes begin to approach the numerical error floor. The $P_2 \times P_0$ discretisation shows lower but still systematic convergence, with velocity and pressure errors converging at approximately second and first order, respectively.

For the Kramer annulus benchmark, the observed convergence behaviour depends primarily on the regularity of the forcing term. The smooth volumetric forcing cases recover close to the expected rates for the $P_2 \times P_1$ discretisation, with pressure errors converging at approximately second order and velocity errors showing second-order or better convergence. This behaviour is consistent with the results reported by Kramer et al. [2021], where the use of linear geometric meshes limits the velocity L_2-norm convergence to approximately second order. Achieving the ideal third-order

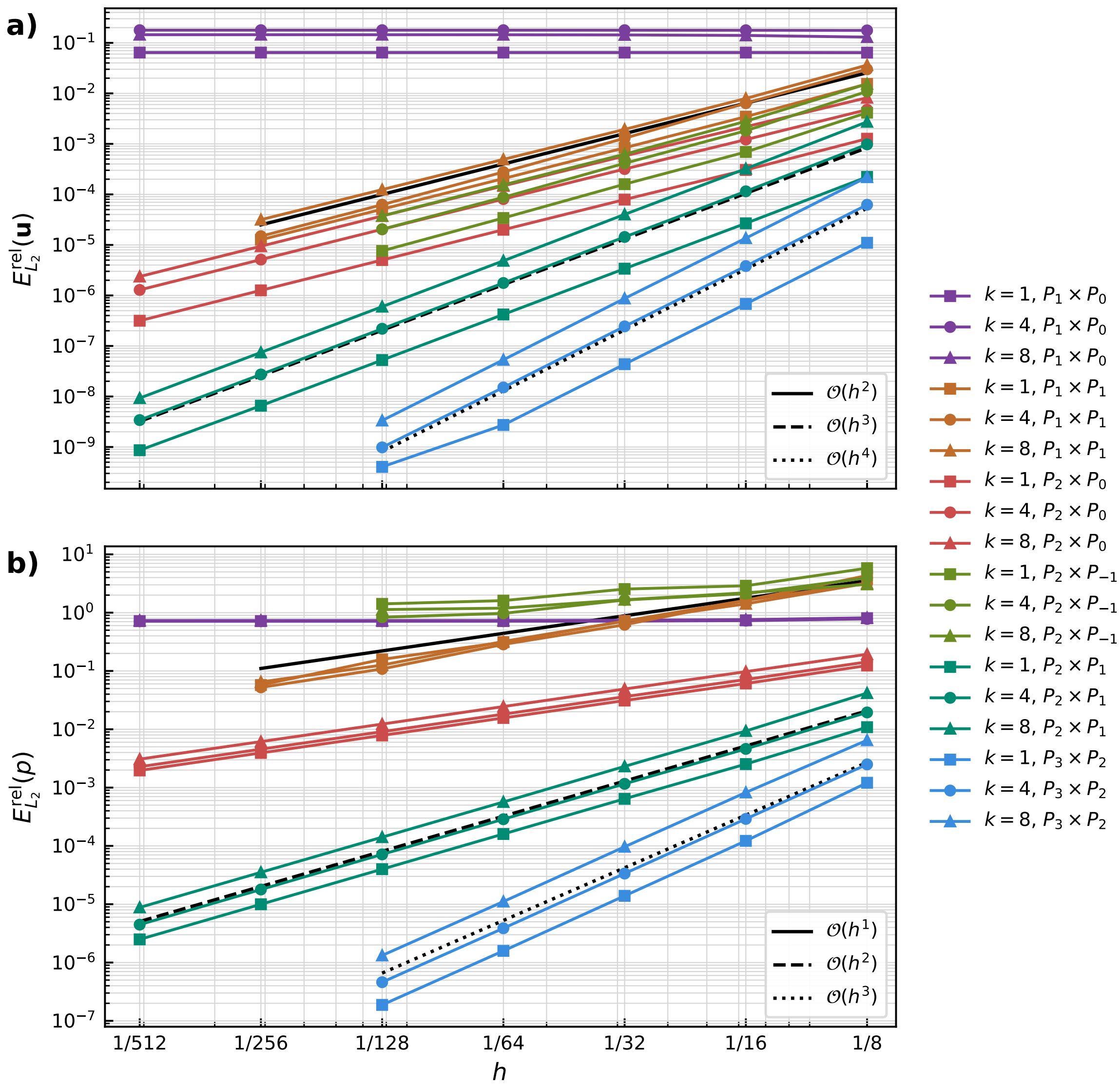

Figure 1: Velocity and pressure convergence for the Thieulot–Puckett annulus benchmark.

velocity convergence for the $P_2 \times P_1$ discretisation requires a higher-order isoparametric approximation of the curved mesh geometry. In contrast, the delta-function forcing cases introduce reduced solution regularity through the internal interface, leading to slower convergence of approximately $O(h^{1.5})$ for velocity and $O(h^{0.5})$ for pressure. These reduced rates are expected from the analytical structure of the benchmark and do not indicate a deficiency in the Stokes solver.

The three-dimensional Thieulot spherical-shell benchmark provides the clearest smooth-solution verification case in spherical geometry. For the $P_2 \times P_1$ Taylor–Hood discretisation, both the constant-viscosity and radially varying viscosity models approach $O(h^3)$ convergence in velocity and $O(h^2)$ convergence in pressure. These results confirm that the spherical-shell Stokes implementation preserves the expected mixed finite-element convergence hierarchy on curved three-dimensional domains.

The Kramer spherical-shell benchmark exhibits the same distinction between smooth and singular forcing observed in the annulus configuration. The smooth forcing cases produce pressure convergence close to $O(h^2)$, while the velocity errors converge at rates closer to $O(h^2)$ than to the ideal P_2 L_2-norm rate over the tested mesh sequence. The delta-function forcing cases again recover the expected reduced convergence trends of approximately $O(h^{1.5})$ in velocity and $O(h^{0.5})$ in pressure. Overall, the annulus and spherical-shell benchmark suites demonstrate that the Stokes implementation reproduces the expected convergence behaviour for smooth analytical solutions and the anticipated reduction in convergence rates when the forcing contains singular internal interfaces.

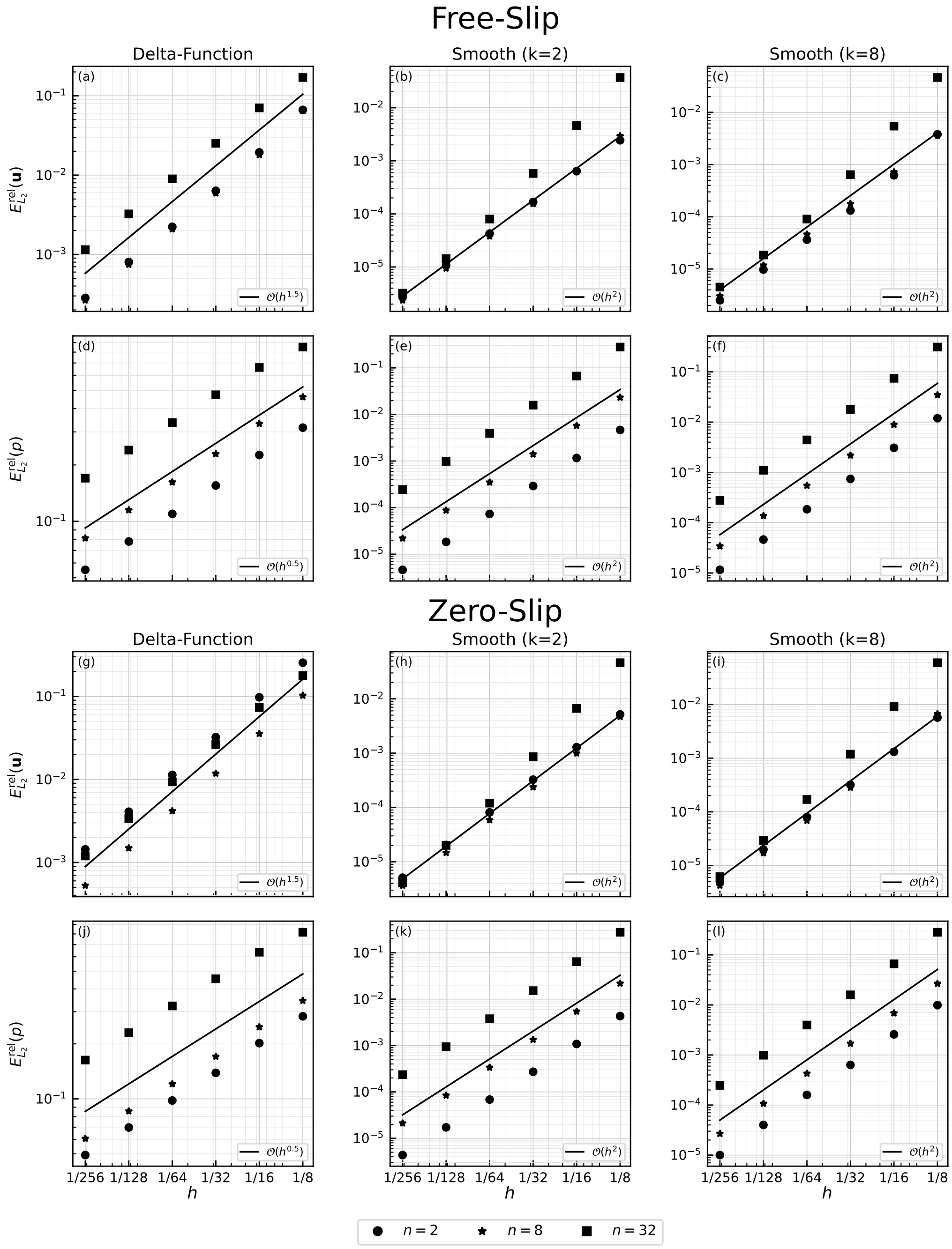

Figure 2: Velocity and pressure convergence for the Kramer annulus benchmark.

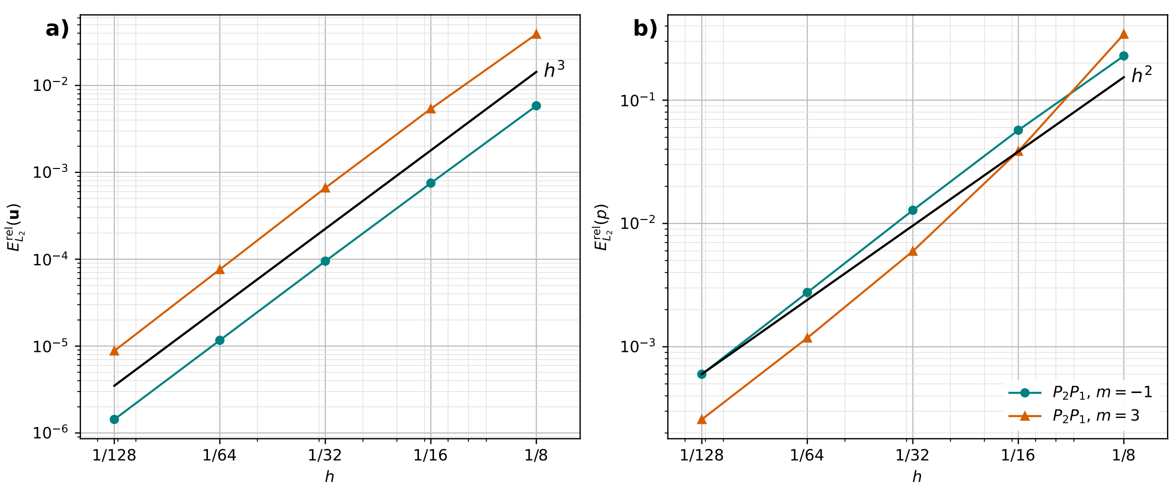

Figure 3: Velocity and pressure convergence for the Thieulot spherical-shell benchmark.

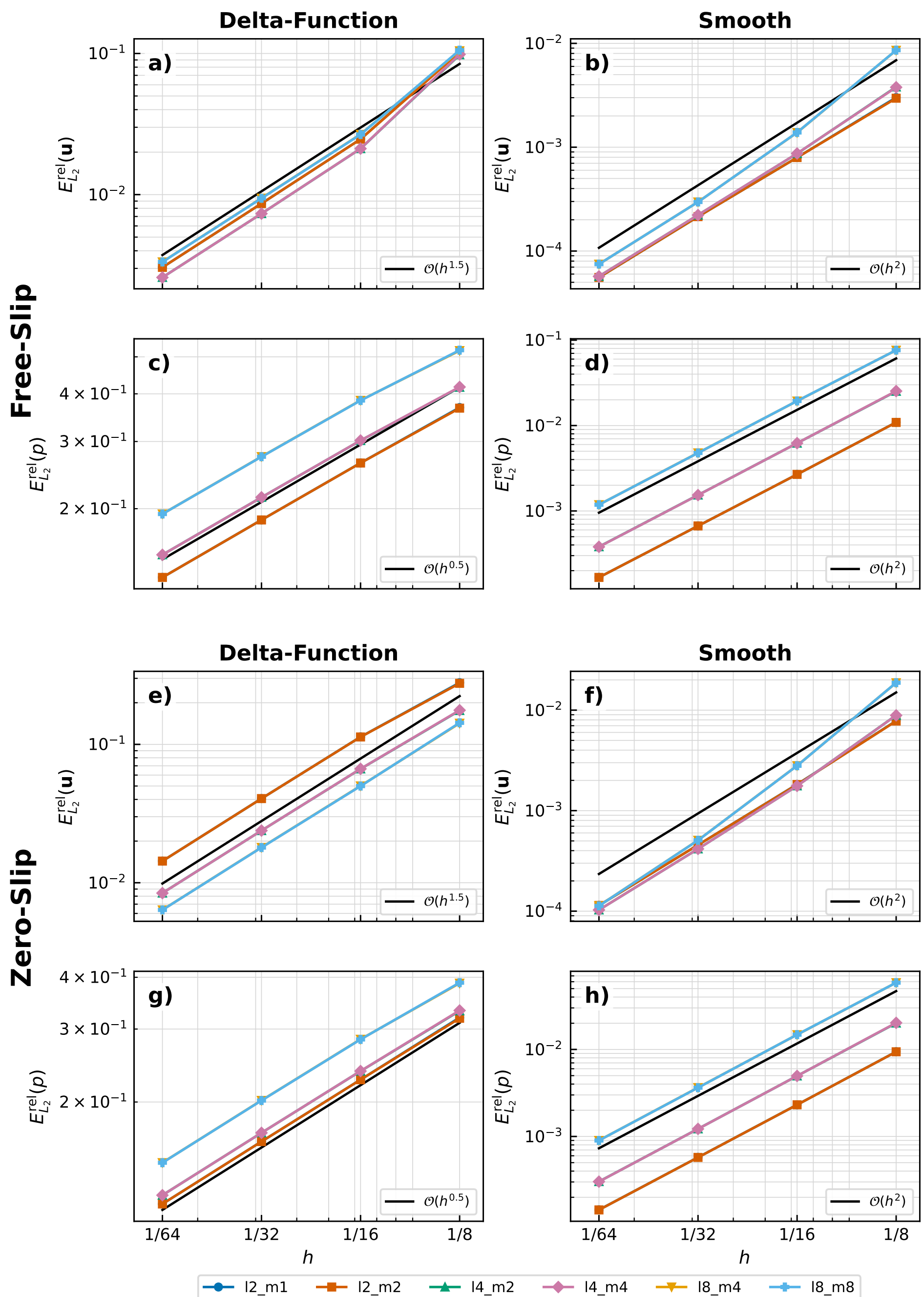

Figure 4: Velocity and pressure convergence for the Kramer spherical-shell benchmark.

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